

URBAN HEAT ISLAND STUDY

across Indian Metro Cities

Problem Statement

Rapid urbanisation in Indian metro cities has intensified the Urban Heat Island (UHI) effect, where urban areas experience higher temperatures than surrounding rural regions due to impervious surfaces, reduced vegetation, loss of water bodies, and anthropogenic heat emissions. This study aims to analyse and compare Surface Urban Heat Island (SUHI) patterns across major Indian metropolitan cities using multi-source satellite data and machine learning techniques. The objective is to identify spatial heat hotspots, quantify daytime and nighttime UHI intensity, and determine the key environmental and urban factors driving urban warming to support sustainable and climate-resilient urban planning.

Git link: https://github.com/Divyareddy8/IndiHeat_ML

Collab file for delhi(.ipynb given in git also) :

<https://colab.research.google.com/drive/1KeM3nkvWHbPNJexQD1QHVGgGauOh5moW?usp=sharing>

Divya - IMT2023059

Surface Urban Heat Island (SUHI) Analysis

Delhi · Mumbai · Bengaluru · Chennai · Kolkata · Hyderabad · Ahmedabad · Pune

Study Period: 2019–2023 | Platform: Google Earth Engine

1. Study Overview

This report presents the methodology adopted for the Urban Heat Island (UHI) analysis of Delhi, the first city in a broader multi-city study of Surface Urban Heat Islands (SUHI) across major Indian metropolitan regions. The wider study includes Mumbai, Bengaluru, Chennai, Kolkata, Hyderabad, Ahmedabad, and Pune, selected to represent diverse urban climates, development patterns, and geographic settings across India.

Each city contributes a distinct urban context to the study:

- **Delhi** — severe heat waves, dense impervious surfaces, and strong nighttime UHI effects.
- **Mumbai and Chennai** — representative coastal tropical cities influenced by humidity and marine interactions.
- **Bengaluru and Pune** — rapidly expanding plateau cities experiencing significant urban growth and land-use change.
- **Kolkata** — dense humid eastern metropolitan environment with strong moisture influence.
- **Hyderabad** — semi-arid Deccan urbanisation with pronounced land surface temperature gradients.
- **Ahmedabad** — extreme heat conditions and a nationally recognised heat-action planning framework.

Delhi was selected as the starting point because it is one of the most extensively studied Indian cities in UHI research and exhibits a strong thermal signature driven by rapid urbanisation and extensive built-up surfaces. The methodology developed in this report serves as a consistent and replicable framework for subsequent analyses across the remaining cities.

Study Area	National Capital Territory (NCT) of Delhi — approximately 1,484 km²
Seasons Analysed	Summer (April–June) and Winter (December–February)
Time of Day	Daytime (~10:30 AM local) and Nighttime (~10:30 PM local)
Study Period	2019–2023 (multi-year median composites)
Platform	Google Earth Engine (for feature extraction) & QGIS for visualizations

2. Data Sources

The analysis draws on multiple satellite and geospatial datasets, each selected for a specific purpose.

Satellite Imagery

1. **Landsat 8 and Landsat 9**
Landsat 8 and Landsat 9 Collection 2 Level-2 (USGS) datasets were used for daytime Land Surface Temperature (LST) estimation and computation of spectral indices. Collection 2 Level-2 products are atmospherically corrected and provide Surface Reflectance (SR) and Surface Temperature (ST) products suitable for scientific analysis with minimal additional preprocessing.
2. **MODIS MOD11A1**
MODIS MOD11A1 (Terra, 1 km daily resolution) was used for nighttime LST analysis, capturing the approximately 10:30 PM overpass, when the Urban Heat Island (UHI) signal is typically strongest.
3. **Sentinel-2 SR Harmonised**
Sentinel-2 SR Harmonised imagery was used where required for Local Climate Zone (LCZ) classification, particularly due to its 10 m spatial resolution and enhanced spectral capabilities.
4. **Sentinel-5P TROPOMI**
Sentinel-5P TROPOMI data were used to derive air-quality covariates, including NO₂, CO, and UV Aerosol Index, serving as proxy indicators for traffic intensity and industrial activity.

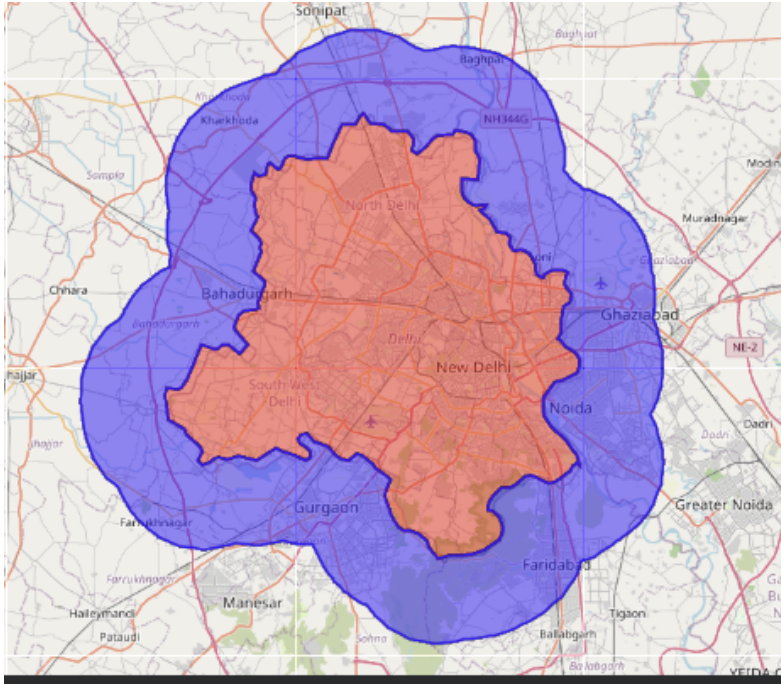
Ancillary Datasets

5. **USGS SRTM 30 m DEM**
USGS SRTM 30 m DEM was used to derive elevation, slope, aspect, and Topographic Wetness Index (TWI).
6. **WorldPop 2020 (100 m)**
WorldPop 2020 population density data were used as a proxy for urban intensity.
7. **ESA WorldCover 2021 (10 m)**
ESA WorldCover 2021 was used for land use/land cover (LULC) mapping, including impervious surface identification.
8. **JRC Global Surface Water**
JRC Global Surface Water data were used to identify permanent water bodies for proximity analysis.
9. **FAO GAUL 2015 Level-1**
FAO GAUL 2015 Level-1 administrative boundaries were used to define the National Capital Territory (NCT) of Delhi as the study region of interest.

3. Methodology

3.1 Region of Interest and Rural Reference Zone

The Delhi NCT boundary was defined from the FAO GAUL Level-1 dataset, covering approximately 1,484 km². A 10 km annular rural buffer was created around the NCT — pixels inside the buffer but outside the NCT boundary serve as the rural reference for UHI intensity calculations. This approach isolates the urban signal from any edge effects at the city boundary.



3.2 Landsat Preprocessing Pipeline

Every Landsat 8/9 scene was passed through a standardised pre-processing pipeline before any analysis:

- Cloud and shadow masking using the QA_PIXEL bit-flag band, removing dilated clouds, cirrus, cloud shadows, and saturated pixels — the USGS-recommended approach for Collection 2 data.
- Scale factor application per USGS documentation: Surface Reflectance bands multiplied by 2.75×10^{-5} and offset by -0.2 ; Surface Temperature band multiplied by 0.00341802 and offset by 149 K.
- Conversion of the thermal band (ST_B10) from Kelvin to degrees Celsius for all subsequent LST analysis.

- Computation of nine spectral indices — NDVI, NDBI, NDWI, MNDWI, SAVI, EVI, UI, BSI, and broadband Albedo — each selected on the basis of established literature as UHI driving factors.

Multi-year median composites (2019–2023) were then generated for Summer (April–June) and Winter (December–February) separately. Median compositing was chosen over mean compositing to reduce the influence of residual cloud contamination and inter-annual anomalies, following the approach of Zhou et al. (2023).

3.3 Nighttime LST from MODIS

Nighttime LST was derived from the MODIS MOD11A1 daily 1 km product. A quality-control filter was applied, retaining only pixels with mandatory QA bits 0–1 equal to 00 (Good) or 01 (Other Quality), discarding cloud-contaminated and unproduced retrievals. The scale factor of 0.02 was applied and values converted to Celsius. Median composites for 2019–2023 were computed for each season. The MODIS native sinusoidal projection was pinned and then reprojected to EPSG:4326 at 1 km for consistency with Landsat-derived layers.

3.4 TROPOMI Sentinel-5P Feature Extraction

Air-quality co-variates were extracted from Sentinel-5P TROPOMI datasets for integration into Urban Heat Island machine learning modelling.

Products Used

- **NO₂ tropospheric column density** (OFFL/L3_N02)
- **CO total column density** (OFFL/L3_CO)
- **UV Aerosol Index** (OFFL/L3_AER_AI)

Science Rationale

The inclusion of atmospheric variables was motivated by their documented relationship with urban thermal conditions:

- **NO₂** acts as a proxy for traffic density and industrial emissions that commonly co-locate with Urban Heat Island hotspots (Bechle et al., 2011).
- **CO** represents combustion-related activity, including industrial emissions, vehicular pollution, and biomass burning.
- **UV Aerosol Index** influences incoming shortwave radiation and atmospheric scattering, thereby affecting surface thermal behaviour and observed Land Surface Temperature.

These variables were incorporated as proxy indicators of anthropogenic activity and atmospheric conditions influencing urban thermal dynamics.

3.5 Driving Factor Extraction (Feature Engineering)

A comprehensive stack of 23 spatial predictor variables was assembled for machine learning modelling. These were selected through a systematic review of 40+ UHI papers from 2020–2024, ensuring each variable has a documented physical or socioeconomic link to urban heat:

- Spectral indices: NDVI, NDBI, NDWI, MNDWI, SAVI, EVI, UI, BSI, Albedo, Emissivity — capturing vegetation cover, built-up intensity, water, and surface optical properties.
- Vegetation Fraction (FV): derived from NDVI using Delhi-specific bare soil and full vegetation thresholds (Gupta et al., 2020), representing fractional canopy cover.
- Topographic variables: Elevation and Slope from SRTM 30 m; Topographic Wetness Index as a proxy for moisture accumulation in low-lying areas.
- Population density (WorldPop 2020) and impervious surface fraction (ESA WorldCover built-up class) — direct indicators of urban intensity.
- Distance to water bodies and distance to green spaces — capturing the cooling influence of blue-green infrastructure.
- Built-up density: focal mean of impervious surface within a 500 m radius, approximating neighbourhood-scale urban compactness.
- Sky View Factor proxy: inverse of focal NDBI, used as an approximation of sky openness in the absence of full 3D morphological data.
- Air-quality co-variables from Sentinel-5P TROPOMI: NO₂ tropospheric column density ($\mu\text{mol}/\text{m}^2$) as a proxy for traffic and industrial emissions; CO total column density (mmol/m^2) as an indicator of combustion activity; UV Aerosol Index as a modifier of incoming shortwave radiation.

3.6 Local Climate Zone (LCZ) Classification

The Stewart and Oke (2012) Local Climate Zone (LCZ) framework was used to divide the city into 17 urban and natural classes, enabling a physically consistent comparison of UHI intensity across different urban forms. For Delhi, the analysis primarily used the Demuzere et al. (2022) global WUDAPT LCZ dataset available in the Google Earth Engine catalogue, a pre-computed 100 m raster that eliminates the need for computationally expensive on-the-fly classification.

Where this global product was not available, a fallback Random Forest model was trained using WUDAPT training polygons and a Sentinel-2 feature stack (NDVI, NDBI, NDWI, elevation, and impervious surface fraction). The model used 50 trees at 200 m resolution to keep computation lightweight while maintaining reliable classification performance.

Urban areas were defined as LCZ classes 1–10 (built-up types), while the rural reference was based on LCZ class 11 (dense trees), chosen as a conservative vegetation baseline to minimise peri-urban influence.

3.7 UHI Intensity Computation

Per-pixel UHI intensity was computed as:

$$\text{UHI}(x, y) = \text{LST}(x, y) - T_{\text{rural}}$$

where T_{rural} is the median LST of all rural reference pixels (LCZ 11) within the ROI. The median was preferred over the mean as the rural baseline because it is more robust to the small number of remaining rural pixels in the densely urbanised Delhi NCT. Four UHI intensity maps were produced — Daytime Summer, Nighttime Summer, Daytime Winter, Nighttime Winter — and mean UHI intensity was then aggregated by LCZ class to produce a stratified comparison table.

```
... Computing Improved UHI maps...
Day Summer: Rural Median=41.48°C | Urban Mean UHI=+0.98°C
Night Summer: Rural Median=22.12°C | Urban Mean UHI=+3.28°C
Day Winter: Rural Median=21.31°C | Urban Mean UHI=-0.13°C
Night Winter: Rural Median=9.47°C | Urban Mean UHI=+2.36°C

Computing LCZ UHI statistics...
Scenario          Day-Summer  Day-Winter  Night-Summer  Night-Winter
LCZ Name
2  Compact Mid-rise    2.37       0.57       5.03       3.66
3  Compact Low-rise   1.53      -0.24       4.31       3.12
4  Open High-rise    -2.10      -0.85       4.01       3.24
5  Open Mid-rise      0.40      -0.17       4.66       3.37
6  Open Low-rise      0.34      -0.29       2.42       1.73
8  Large Low-rise     1.83       0.48       3.09       2.27
9  Sparsely Built      2.49       0.08      -0.57       0.00
11 Dense Trees        -2.05     -1.14       1.91       1.29
12 Scattered Trees    1.29       0.54       0.40      -0.04
13 Bush/Scrub        2.59       1.13       0.04     -0.35
14 Low Plants         3.42     -0.32     -0.73       0.00
15 Bare Rock/Paved    3.79       1.48       3.69       2.37
16 Bare Soil/Sand     4.59       1.16       0.96       0.67
17 Water             -4.12     -1.38       2.63       2.44
```

These values show how different urban forms in Delhi deviate from the surrounding rural baseline in terms of Urban Heat Island (UHI) intensity. Positive values mean the Local Climate Zone (like dense buildings, bare roads, low vegetation) is **warmer than rural areas**, which is common in compact or paved urban zones due to heat trapping and low evapotranspiration.

Negative values indicate zones (like trees or water bodies) that are **cooler than rural reference areas**, because vegetation and water reduce temperature through shading and evaporation.

3.8 Machine Learning Modelling (SHAP-Explained)

A stratified random sample of approximately 100 points per LCZ class (~900 total for Delhi's observed classes) was drawn at 300 m resolution from the full feature stack and LST target. Three models were trained and evaluated:

1. **Random Forest Regressor** — an ensemble of decision trees providing strong baseline performance and native feature importance.
2. **XGBoost** — gradient-boosted trees with regularisation, generally offering higher accuracy and handling non-linear interactions well.
3. **MLP Neural Network** — a multi-layer perceptron to capture complex non-linear patterns in the feature-LST relationship.

All models were evaluated using 5-fold cross-validation with R^2 , RMSE, and MAE. SHAP (SHapley Additive exPlanations) values were computed for the best-performing model to attribute the contribution of each driving factor to individual LST predictions — making the model interpretable and grounding the results in physical understanding rather than treating the ML output as a black box.

Delhi result:

```
Full comparison table:
      Model  R2_Train  R2_Test  RMSE_Test  MAE_Test
Random Forest    0.9437    0.8825    1.0386    0.8020
      XGBoost    0.9777    0.9116    0.9005    0.7037
MLP Neural Net    0.9607    0.8999    0.9587    0.7649
```

Feature Importance order obtain for delhi

```
SHAP Feature Ranking:
01. NDBI           : 1.6714
02. Emissivity     : 0.3695
03. Albedo         : 0.2745
04. BuiltDensity   : 0.1766
05. Elevation      : 0.1737
06. CO             : 0.1699
07. FVC            : 0.1668
08. UVAI           : 0.1330
09. NO2            : 0.1264
10. NDWI           : 0.1218
11. EVI            : 0.0701
12. NDVI           : 0.0651
13. Population     : 0.0539
14. SAVI           : 0.0329
15. Slope          : 0.0282
16. Impervious     : 0.0121
```


4. Rationale for Key Methodological Decisions

Several methodological choices warrant brief justification:

13. Multi-year median composites instead of single-year maps:

Delhi's monsoon season leads to persistent cloud contamination, often causing gaps in thermal observations. Using median composites from 2019–2023 ensures spatial completeness while reducing the influence of interannual anomalies, including atypical conditions during the 2020 COVID-19 lockdown period.

14. Use of separate daytime (Landsat) and nighttime (MODIS) datasets:

No single satellite sensor provides both daytime and nighttime land surface temperature at suitable spatial resolution. Landsat thermal data (30 m) enables fine-scale spatial analysis during daytime, while MODIS offers near-daily revisit capability at ~1 km resolution, making it the standard dataset for nighttime UHI characterization.

15. LCZ-based rural reference instead of a fixed buffer:

Simple peri-urban buffer approaches may inadvertently include heterogeneous land uses such as industrial zones or mixed urban fringes, biasing rural temperature estimates. Using LCZ 11 (Dense Trees) as a reference provides a physically meaningful, standardized, and reproducible baseline for inter-city comparison.

16. Inclusion of TROPOMI air-quality variables:

These covariates enable the model to capture indirect effects of atmospheric composition on UHI, particularly aerosol loading and emission intensity. This relationship is well established in the literature but remains underrepresented in urban heat studies over Indian cities.

17. SHAP-based interpretability:

In a multi-city comparative framework, the primary objective extends beyond prediction of land surface temperature to understanding the drivers of UHI variability. SHAP values ensure model interpretability and allow consistent comparison of feature importance across cities.

5. City-by-City Results

5.1 Delhi — National Capital Territory (NCT)

Delhi was selected as the primary study city due to its well-documented thermal signature, dense impervious surfaces, and one of the strongest UHI signals in South Asia. The NCT covers approximately 1,484 km² and was analysed across daytime and nighttime seasons using Landsat 8/9 and MODIS data.

Daytime & Night LST Stats:

```
Daytime LST Statistics (°C):  
Summer Day:  
  LST_Day_Summer_max: 50.66  
  LST_Day_Summer_mean: 43.24  
  LST_Day_Summer_min: 32.85  
  LST_Day_Summer_stdDev: 2.51  
Winter Day:  
  LST_Day_Winter_max: 25.97  
  LST_Day_Winter_mean: 21.12  
  LST_Day_Winter_min: 17.93  
  LST_Day_Winter_stdDev: 0.92
```

```
Night LST Stats:  
Summer Night : {'LST_Night_Summer_max': 28.084285714285727, 'LST_Night_Summer_mean': 24.066477980460686, 'LST_Night_Summer_min': 18.900000000000034}  
Winter Night : {'LST_Night_Winter_max': 13.970000000000027, 'LST_Night_Winter_mean': 11.053368868863918, 'LST_Night_Winter_min': 7.503333333333333}
```

Daytime UHI

The daytime map clearly reveals intense heat cores in south-west Delhi (Dwarka–Najafgarh axis) and north-west industrial corridors, reaching temperatures 6–9°C above the rural baseline. The Yamuna floodplain and Ridge Forest appear as distinct cool zones, demonstrating the strong cooling influence of water bodies and dense tree cover.

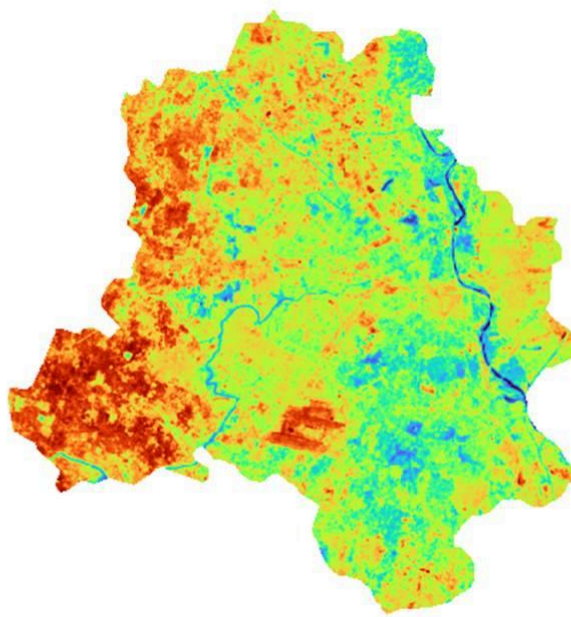


Figure 1.1 — Delhi: Daytime Summer UHI (Landsat 8/9 Composite, 2019–2023)

Key Reasons for Daytime UHI in Delhi

1. **Impervious urban surfaces:** Concrete, asphalt, and rooftops absorb and store large amounts of solar radiation, causing higher daytime temperatures.
2. **Low vegetation cover:** Reduced greenery decreases evapotranspiration and shading, leading to stronger surface heating.
3. **Anthropogenic heat emissions:** Traffic, industries, and air-conditioning systems add extra heat to the urban environment.

Predicted LST — Mean: 43.26°C | Std: 2.36°C
Rural reference LST (LCZ 11 median): 37.90°C
Predicted UHI — Mean: 5.30°C | Max: 11.69°C

Vegetation Cover (NDVI)

NDVI values are uniformly low across the built-up core of Delhi, confirming the vegetation deficit that drives daytime surface heating. Greener zones along the Ridge Forest and Yamuna floodplain correlate spatially with the cooler LST clusters seen above.

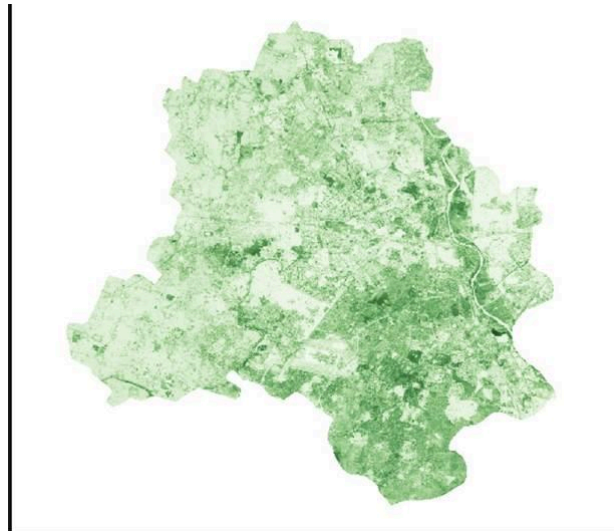


Figure 1.2 — Delhi: NDVI (Vegetation Cover Map, 2019–2023 Median)

Impervious Surface (Built-Up Density)

High impervious surface fractions dominate the western and southern districts. The spatial correspondence between dense built-up areas and high LST zones is direct, confirming impervious surface as the primary daytime UHI driver in Delhi.

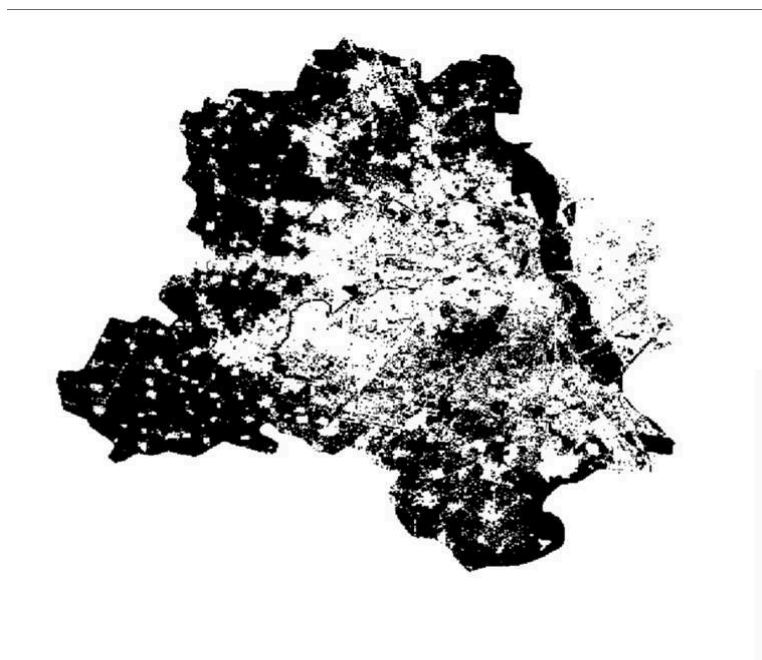


Figure 1.3 — Delhi: Impervious Surface / Built-Up Density (ESA WorldCover)

Air Quality Co-variate (NO₂ / Aerosol)

Elevated NO₂ concentrations (yellow–green zones) in the central and eastern urban core act as proxy indicators of traffic intensity and industrial combustion. High aerosol loading modifies incoming shortwave radiation and contributes indirectly to the UHI signal through reduced nighttime radiative cooling.

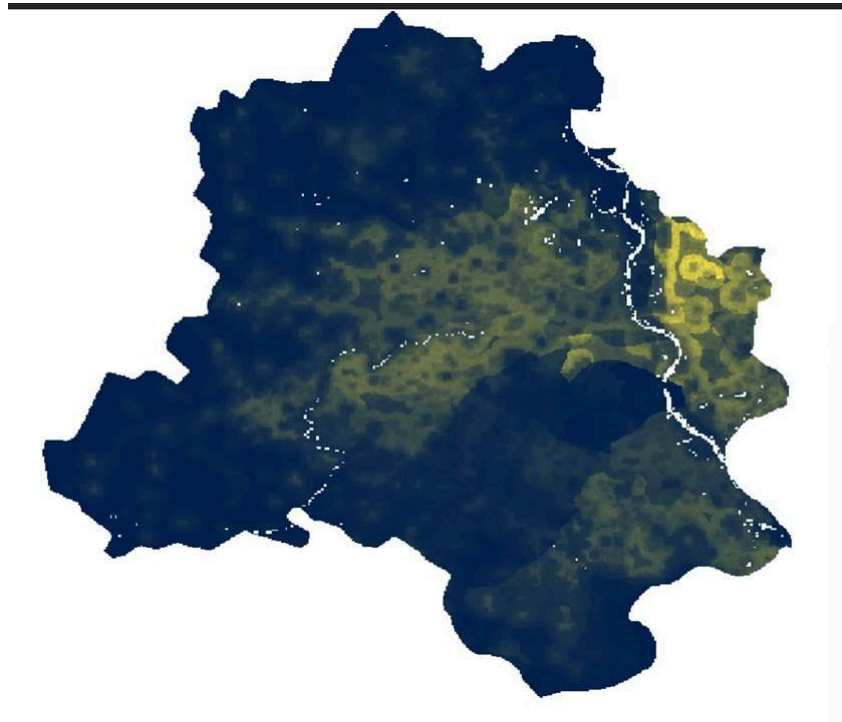


Figure 1.4 — Delhi: Tropospheric NO₂ Column Density (Sentinel-5P TROPOMI)

Nighttime UHI

Nighttime temperatures remain elevated across the central urban mass, with the strongest residual heat trapped in compact residential and commercial zones. The Yamuna corridor and peripheral agricultural areas cool faster, producing clear urban–rural thermal contrasts even at 10:30 PM.

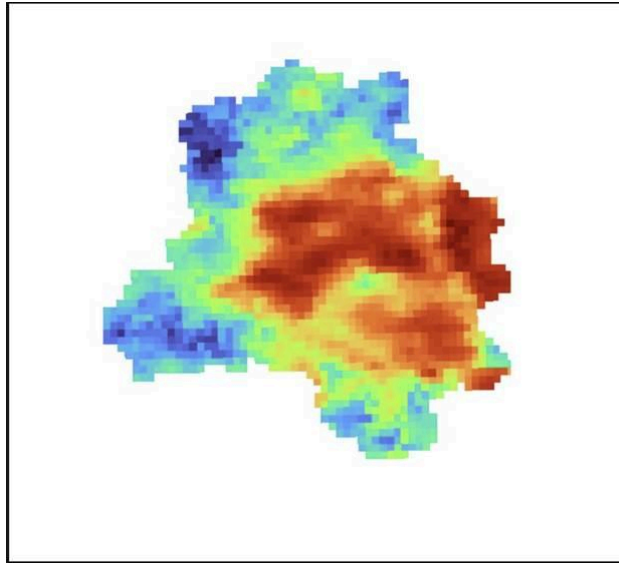


Figure 1.5 — Delhi: Nighttime Summer UHI (MODIS MOD11A1 Composite, 2019–2023)

Key Reasons for Nighttime UHI in Delhi

4. **Thermal inertia of urban materials:** Concrete and asphalt accumulate heat through the day and release it slowly at night, sustaining elevated surface temperatures well into the early morning hours.
5. **Urban canyon effect:** Delhi's dense mid-rise fabric traps outgoing longwave radiation within street canyons, preventing effective radiative cooling.
6. **Reduced nighttime ventilation:** Compact building clusters limit the sky view factor, restricting the area from which urban surfaces can radiate heat away to the atmosphere.

Supplementary Spatial Layers

Three additional spatial layers — Local Climate Zone classification, built-up density, and distance to green spaces — were used in the machine learning feature stack for Delhi and are shown below for spatial context.

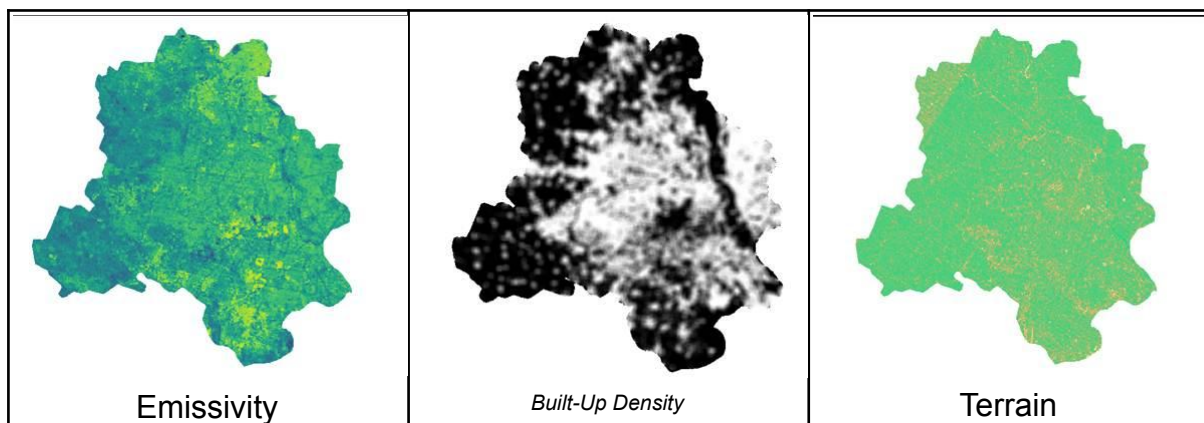


Figure 1.6 — Delhi: Supplementary spatial layers used in ML feature stack (2019–2023)

5.2 Mumbai — Coastal Tropical Metropolis

Mumbai presents a coastal UHI pattern strongly modulated by the Arabian Sea to the west and Thane Creek to the east. Despite high humidity suppressing peak daytime temperatures relative to inland cities, the dense informal settlements and reclaimed land in central and eastern Mumbai generate significant localised heat anomalies.

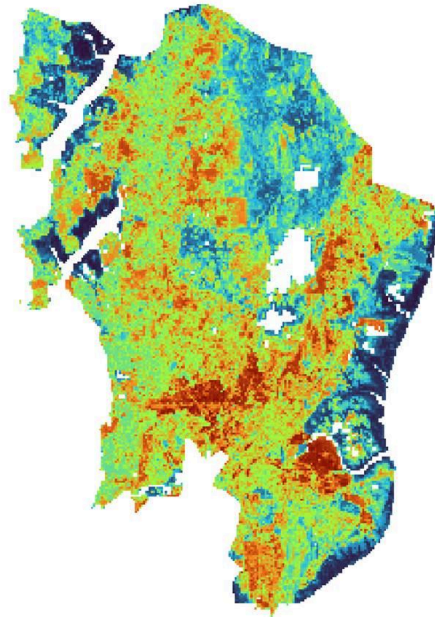


Figure 2.1 — Mumbai: Pre-Monsoon UHI (Landsat 8/9 Composite, 2019–2023)

The map shows intense heating over the dense mid-city zones of Dharavi, Kurla, and Govandi, while the western coastal fringe and creek water bodies remain cool. The sprawling eastern suburbs display sustained warmth due to low vegetation and high impervious fractions despite proximity to water.

Key Reasons for UHI in Mumbai

1. **High-density informal built fabric:** Dense low-rise slum settlements with metal roofing and minimal vegetation create intense localised heat pockets, particularly in Dharavi and Mankhurd.
2. **Loss of mangrove and tidal wetlands:** Historical reclamation of wetland areas has removed a significant natural cooling buffer, increasing surface temperatures in the eastern suburbs.
3. **Sea-breeze disruption by high-rises:** The dense wall of high-rise construction along the western coast increasingly blocks the moderating sea breeze from penetrating inland, reducing its cooling effect on the city core.

5.3 Chennai — Coastal South Indian Metro

Chennai experiences a bimodal rainfall pattern (north-east and south-west monsoons) and high ambient humidity year-round. The UHI effect is driven strongly by the inland industrial belt and is partially moderated along the coast by the Bay of Bengal. Pre-monsoon summers are the most thermally stressed period.

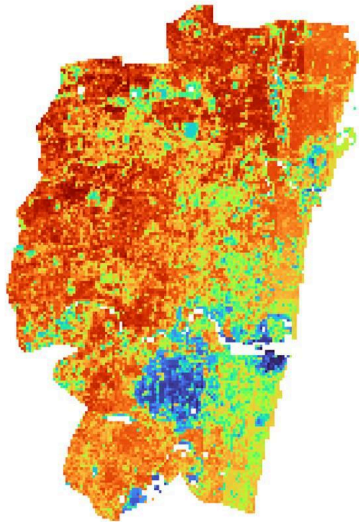


Figure 3.1 — Chennai: Pre-Monsoon UHI (Landsat 8/9 Composite, 2019–2023)

The northern and north-western industrial zones (Ambattur, Padi) record the highest LST values. The southern zone shows cooler conditions where recent tech park development has retained some green cover. A pronounced cool zone in the south corresponds to the Pallikaranai Marshland — one of the few surviving urban wetlands.

Key Reasons for UHI in Chennai

1. **Industrial concentration in the north-west:** Ambattur and Padi industrial estates are among the densest industrial clusters in South India, generating high anthropogenic heat alongside extensive impervious surfaces.
2. **Rapid conversion of tank and wetland land:** Chennai historically had hundreds of water retention tanks that cooled surrounding areas; large-scale encroachment has eliminated most of these, removing their evaporative cooling function.
3. **Dark rooftop materials:** The prevalence of asbestos and dark-sheet roofing in low-income residential areas amplifies daytime solar absorption, creating micro-scale heat hotspots across the urban fabric.

5.4 Hyderabad — Semi-Arid Deccan Urbanisation

Hyderabad sits on the Deccan Plateau at approximately 500 m elevation, giving it a semi-arid climate with very high pre-monsoon temperatures. Rapid IT-led urbanisation in the western and north-western corridors has dramatically altered the landscape over the study period.

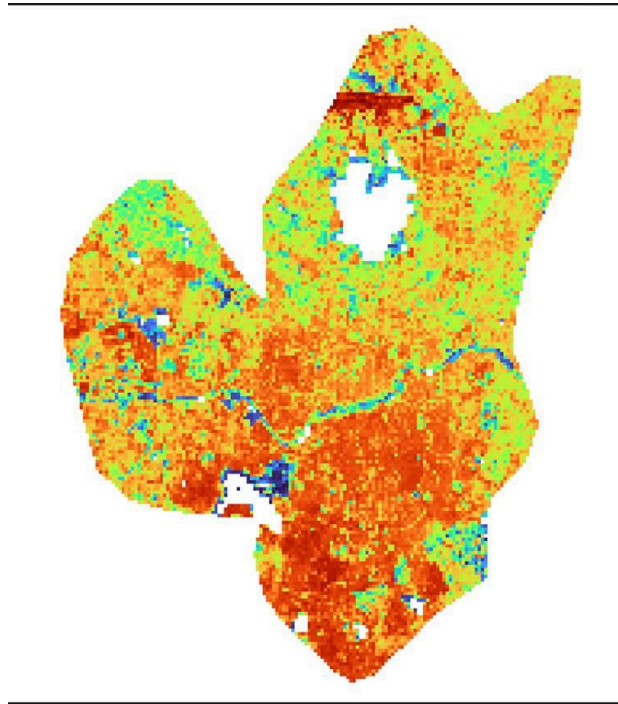


Figure 4.1 — Hyderabad: Pre-Monsoon UHI (Landsat 8/9 Composite, 2019–2023)

The map reveals extensive warming across most of the urban footprint, with the densest heat concentrations in the HITEC City and Gachibowli IT corridors. Hussain Sagar and smaller water bodies appear as localised cool patches, while peripheral agricultural zones remain relatively cooler.

Key Reasons for UHI in Hyderabad

1. **Semi-arid baseline and low-albedo rock:** The Deccan granite bedrock has a naturally low albedo; when built over with concrete, it creates extremely high heat absorption in the already hot pre-monsoon climate.
2. **Loss of Hyderabad's tank network:** The city historically maintained over 3,000 small water tanks; most have been encroached upon, removing a critical evaporative cooling system that kept pre-monsoon temperatures in check.
3. **High-rise construction channelling wind:** The dense IT corridor along the HITEC City and Gachibowli axis has created wind-blocking built masses that reduce natural ventilation and trap anthropogenic heat from data centre operations and vehicular traffic.

5.5 Ahmedabad — Extreme Heat and Heat-Action Planning

Ahmedabad experiences some of the most extreme pre-monsoon temperatures in India, with the city having suffered catastrophic heat mortality events in 2010. It was among the first Indian cities to adopt a formal Heat Action Plan, making it a nationally significant case study for urban heat governance.

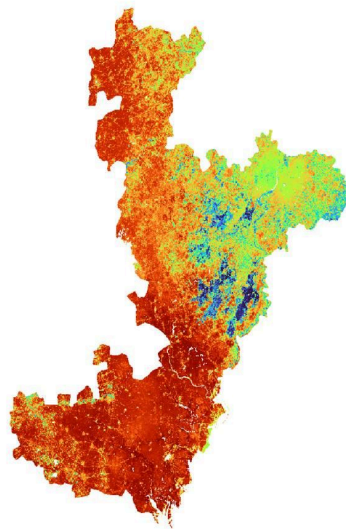


Figure 5.1 — Ahmedabad: Pre-Monsoon UHI (Landsat 8/9 Composite, 2019–2023)

The LST map shows overwhelmingly high surface temperatures across the urban core, with a pronounced heat signature in the densely built western and southern districts. The Sabarmati riverfront corridor provides a narrow linear cool zone. The north-eastern agricultural fringe retains lower temperatures owing to higher vegetation cover and soil moisture.

Key Reasons for UHI in Ahmedabad

4. **Extreme ambient heat baseline:** Ahmedabad's hot semi-arid climate means the baseline temperature is already extreme during pre-monsoon; even modest urban warming from impervious surfaces creates life-threatening heat stress.
5. **Industrial zone expansion:** Rapid industrial growth in the GIDC (Gujarat Industrial Development Corporation) zones generates sustained anthropogenic heat and concentrates impervious surfaces at the urban fringe.
6. **High urban canyon density in the walled city:** Dense old-city walled neighbourhoods (pol areas) with narrow street canyons and high building-to-street ratios trap shortwave radiation and limit nocturnal radiative cooling.
7. **Reduction of the Sabarmati floodplain buffer:** The progressive drying and encroachment of the Sabarmati River floodplain upstream of the riverfront development has reduced a significant evaporative cooling corridor.

5.6 Kolkata — Dense Humid Eastern Metropolis

Kolkata is one of the most densely populated cities in the world and is located in a humid subtropical environment strongly influenced by the Bay of Bengal. High humidity moderates peak daytime LST relative to inland arid cities, but the combination of dense urban fabric, limited green space, and extensive informal settlements produces a persistent and pronounced UHI effect.

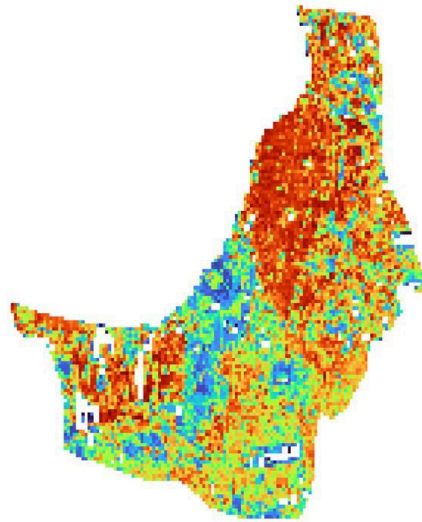


Figure 6.1 — Kolkata: Pre-Monsoon UHI (Landsat 8/9 Composite, 2019–2023)

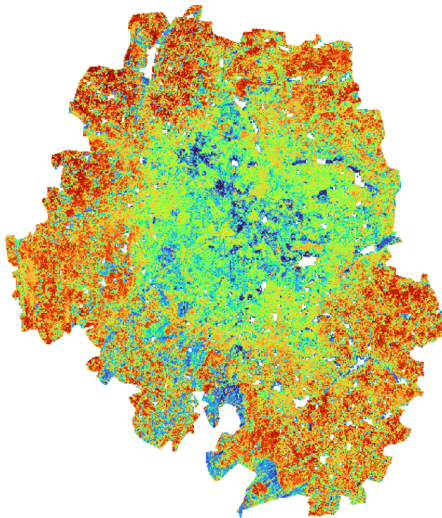
The map reveals intense heating in the central and northern urban core (Shyambazar, Ultadanga, Topsia), where building density is highest. The eastern fringe, including East Kolkata Wetlands (a Ramsar site), shows distinctly cooler temperatures, demonstrating the critical cooling role of wetland ecosystems. The Hooghly River corridor acts as a local thermal moderator on the western edge.

Key Reasons for UHI in Kolkata

8. **Extremely high built density:** Kolkata's urban fabric is characterised by extremely high plot coverage and limited setbacks, resulting in minimal exposed soil and almost no evapotranspiration in the city core.
9. **Encroachment of East Kolkata Wetlands:** Rapid conversion of the eastern wetlands fringe for residential and industrial development is progressively eliminating the most effective natural cooling system in the urban area.
10. **High anthropogenic heat from informal activity:** The city's high population density and informal cooking, transport, and industrial activity contribute substantial waste heat to the urban boundary layer.
11. **High humidity amplifying heat stress:** Seasonal heat stress is compounded by extremely high ambient humidity, which limits evaporative cooling and increases physiological heat stress beyond what LST alone indicates.

5.7 Bengaluru — Rapidly Expanding Plateau City

Bengaluru (Bangalore) sits at approximately 920 m elevation on the Deccan Plateau, historically earning it the reputation of a "Garden City" with a mild, pleasant climate. However, rapid IT-led urbanisation since the 1990s has dramatically transformed its land cover, shrinking its celebrated lakes and green spaces and generating an intensifying urban heat signature.



Key Reasons for UHI in Bengaluru

12. **Severe loss of lake ecosystem:** Bengaluru has lost over 80% of its historical lake extent to encroachment and sewage inflow since the 1970s. These water bodies were the city's primary passive cooling mechanism, and their loss has directly driven LST increases across formerly cooler neighbourhoods.
13. **IT corridor densification:** The IT corridor stretching from Electronic City in the south to Whitefield in the east has generated extremely high-density commercial and residential development with limited green infrastructure, creating sustained heat pockets.
14. **Depletion of residential tree canopy:** Bengaluru's original tree cover in residential layouts (particularly in central areas like Indiranagar and Jayanagar) provided meaningful urban cooling; aggressive construction activity and road widening have progressively removed this canopy.
15. **Elevated solar irradiance at plateau altitude:** At 920 m elevation, Bengaluru's atmosphere is thinner and receives higher solar irradiance than lower-altitude coastal cities, amplifying the heating effect of impervious surfaces during pre-monsoon months.

5.8 Pune — Expanding Plateau City

Pune shares much of its climatic context with Bengaluru — a Deccan Plateau setting, a historically moderate climate, and rapid expansion driven by IT and manufacturing growth. The Pune Metropolitan Region has seen some of the fastest urban growth in India over the past two decades, with significant northward and eastward expansion onto previously agricultural land.

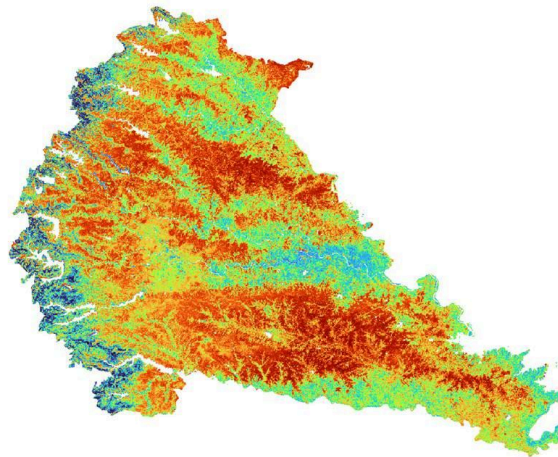


Figure 8.1 — Pune: Pre-Monsoon UHI (Landsat 8/9 Composite, 2019–2023)

The map reveals high surface temperatures across the dense urban core (Hadapsar, Kharadi, Pune cantonment), while the Western Ghats foothills along the western edge of the study area appear distinctly cooler due to forested terrain and elevation. The Mula-Mutha river corridor provides a linear cooling zone through the urban centre.

Key Reasons for UHI in Pune

16. **Agricultural land conversion to urban use:** The conversion of the Mula-Mutha and Pavana river floodplains from agricultural land to industrial estates and residential developments has removed significant evaporative cooling surfaces.
17. **IT and industrial belt in the east:** Pune's eastern expansion — particularly the Hadapsar and Kharadi IT/manufacturing belt — has created large areas of impervious roofing and paving with minimal accompanying green infrastructure.
18. **Unplanned peripheral expansion:** Rapid peripheral growth often occurs through informal and semi-formal development where road, drainage, and green infrastructure standards are not enforced, resulting in exceptionally high impervious coverage.
19. **Emerging canyon effect along arterials:** Like Bengaluru, Pune benefits naturally from its plateau altitude for airflow; however, increasing building heights along major arterial corridors are beginning to obstruct traditional wind channels.

6. Comparative Summary Across Indian Metro Cities

The eight cities studied represent a diverse set of Indian urban typologies, climate zones, and development trajectories. Despite these differences, several cross-cutting drivers of the Urban Heat Island (UHI) effect emerge consistently across all study areas.

City	Climate Type	Primary UHI Drivers	UHI Intensity
Delhi	Semi-arid / continental	Impervious surfaces, dense built fabric	Very High (6–9°C)
Mumbai	Coastal tropical	Informal settlements, loss of wetlands	Moderate–High
Chennai	Coastal humid	Industrial clusters, tank encroachment	High
Hyderabad	Semi-arid Deccan	Low-albedo rock, tank loss, IT corridors	High
Ahmedabad	Hot semi-arid	Extreme heat baseline, industrial zones	Very High
Kolkata	Humid subtropical	Dense built fabric, wetland loss	High
Bengaluru	Plateau semi-humid	Lake loss, IT corridor densification	High
Pune	Plateau semi-arid	Agricultural conversion, peripheral sprawl	Moderate–High

Across all eight cities, the loss of blue-green infrastructure — lakes, tanks, wetlands, mangroves, and river floodplains — emerges as a critical shared driver of intensifying UHI effects. Impervious surface expansion, driven by residential, commercial, and industrial development, compounds this loss by dramatically increasing solar heat absorption and storage. Machine learning modelling with SHAP interpretability will quantify the relative contribution of each driver city-by-city in the final comparative analysis.

All the results are quantitative results are provided in the codes across all the cities